

NAKHON SAWAN, Thailand

WMO: 484000

Lat: 15.6717N

Lon: 100.1322E

Elev: 116

StdP: 14.63

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	61.4	64.2	52.0	57.8	72.6	55.1	64.8	73.5	10.0	83.0	8.7	83.1	0.1	40	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.0	102.2	79.5	100.2	78.9	98.3	78.8	82.9	93.6	82.4	92.8	81.8	91.9	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	159.3	87.8	79.9	156.0	87.1	79.4	153.2	86.5	47.0	94.0	46.3	93.2	45.8	92.2	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.9	9.0	7.5	DB	56.8	104.5	3.8	2.0	54.0	105.9	51.8	107.1	49.6	108.3	46.8	109.8	
			WB	54.1	84.3	3.7	0.9	51.4	85.0	49.3	85.5	47.2	86.0	44.5	86.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.1	79.0	83.4	87.2	89.4	87.5	86.2	85.0	84.5	83.9	83.4	81.5	78.3
	DBStd	4.51	4.09	4.12	3.85	3.55	3.48	2.05	2.05	1.75	1.78	2.07	3.17	4.28
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12447	899	935	1154	1183	1162	1085	1084	1071	1016	1036	944	878
	CDD65	6972	434	515	689	733	697	635	619	606	566	571	494	413
	CDH74	83039	4310	6080	9224	10426	9141	7909	7362	6978	6255	6330	5117	3907
(13)	CDH80	40246	1909	3103	5197	6301	4900	3796	3262	2941	2497	2568	2149	1623
(14)	Wind	WSAvg	1.7	1.3	2.3	3.2	3.0	1.9	2.0	1.9	1.5	0.8	0.7	1.0
(15) Precipitation	PrecAvg	45.9	0.2	0.5	1.6	2.6	6.1	5.3	5.6	6.8	9.8	6.3	0.9	0.2
	PrecMax	61.3	1.0	2.7	6.7	7.7	13.1	10.0	10.0	14.4	14.8	21.9	5.0	2.0
	PrecMin	25.0	0.0	0.0	0.0	0.2	1.4	1.2	1.8	2.6	3.0	0.0	0.0	0.0
	PrecStd	8.6	0.3	0.7	1.6	2.1	3.0	2.2	2.1	2.7	3.0	3.9	1.1	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.2	99.3	102.3	105.7	104.5	99.2	97.2	95.5	94.7	94.5	95.2	94.5
		MCWB	75.6	75.6	77.4	80.6	81.1	80.3	80.6	80.4	81.6	80.5	78.8	76.8
	2%	DB	93.3	97.1	100.3	103.1	100.8	96.3	94.6	93.8	93.0	92.7	93.2	92.4
		MCWB	75.1	76.3	77.2	79.7	80.5	80.5	80.6	80.4	81.1	80.1	78.5	76.2
	5%	DB	91.5	95.3	98.6	101.1	98.1	94.5	92.9	92.3	91.6	91.6	91.5	90.6
		MCWB	74.4	76.3	77.3	79.6	80.5	80.6	80.3	80.3	80.8	79.7	77.8	75.3
	10%	DB	89.5	93.3	96.7	98.9	95.7	92.7	91.4	90.7	90.0	90.2	89.8	88.4
		MCWB	73.6	75.8	77.3	79.6	80.6	80.5	80.0	80.0	80.5	79.6	77.2	74.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	81.0	81.7	83.5	83.9	83.7	82.7	82.7	83.1	82.7	82.0	80.3
		MCDB	89.8	93.4	95.6	98.3	95.8	92.9	92.1	91.2	91.6	91.1	90.8	89.8
	2%	WB	78.4	79.8	80.7	82.4	82.7	82.6	81.8	81.8	82.1	81.8	80.8	78.3
		MCDB	87.9	91.9	94.4	96.6	94.7	92.0	91.0	90.4	90.3	89.8	89.2	87.9
	5%	WB	77.1	78.8	79.9	81.6	81.9	81.8	81.1	81.1	81.5	81.1	79.8	77.1
		MCDB	86.5	90.5	92.8	95.2	93.4	91.1	90.2	89.6	89.3	88.7	87.7	86.5
	10%	WB	75.8	77.8	79.1	80.9	81.3	81.0	80.5	80.5	80.9	80.5	78.7	75.9
		MCDB	84.9	88.6	91.2	93.8	92.2	90.2	89.4	88.9	88.3	87.6	86.1	85.0
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	21.2	19.6	19.0	15.8	14.2	13.4	13.4	13.1	13.6	16.4	19.8
		MCDBR	23.2	22.0	21.6	21.2	19.1	16.5	15.7	15.3	15.2	15.4	18.0	20.8
		MCWBR	7.7	6.2	5.8	5.6	4.9	5.1	5.1	5.2	5.3	4.9	5.4	6.7
	5% WB	MCDBR	18.6	19.3	19.4	19.2	16.8	14.9	14.3	14.2	14.0	13.9	15.6	18.1
		MCWBR	6.8	5.9	5.7	5.8	5.5	5.5	5.4	5.4	5.5	5.2	5.4	6.1
(40) Clear-Sky Solar Irradiance	taub		0.507	0.595	0.672	0.602	0.497	0.442	0.434	0.438	0.476	0.561	0.495	0.498
	taud		2.010	1.821	1.679	1.877	2.219	2.382	2.405	2.379	2.260	1.973	2.132	2.050
	Ebn at Noon		248	232	219	235	257	269	272	273	264	238	250	246
	Edh at Noon		54	68	80	66	46	39	38	39	44	57	47	50
(44) All-Sky Solar Radiation	RadAvg		1511	1660	1734	1894	1784	1661	1564	1527	1510	1509	1528	1476
(45)	RadStd		90	66	109	89	113	102	91	90	79	88	82	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.76	+0.72	N/A	N/A	+180	+180
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	+0.36	+0.45	N/A	N/A	N/A	N/A

Nomenclature: See separate page